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Eye and Identity ★

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Problem

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identity

The identity tool returns an identity array. An identity array is a square matrix with all the main diagonal elements as **1** and the rest as **0**. The default type of elements is float.

```
import numpy
print numpy.identity(3) #3 is for dimension 3 X 3
```

```
#Output
[[ 1.  0.  0.]
 [ 0.  1.  0.]
 [ 0.  0.  1.]]
```

eye

The eye tool returns a 2-D array with **1**'s as the diagonal and **0**'s elsewhere. The diagonal can be main, upper or lower depending on the optional parameter **k**. A positive **k** is for the upper diagonal, a negative **k** is for the lower, and a **0 k** (default) is for the main diagonal.

```
import numpy
print numpy.eye(8, 7, k = 1) # 8 X 7 Dimensional array with first upper diagonal
```

```
#Output
[[ 0.  1.  0.  0.  0.  0.  0.]
 [ 0.  0.  1.  0.  0.  0.  0.]
 [ 0.  0.  0.  1.  0.  0.  0.]
 [ 0.  0.  0.  0.  1.  0.  0.]
 [ 0.  0.  0.  0.  0.  1.  0.]
 [ 0.  0.  0.  0.  0.  0.  1.]
 [ 0.  0.  0.  0.  0.  0.  0.]
 [ 0.  0.  0.  0.  0.  0.  0.]]
```

```
print numpy.eye(8, 7, k = -2) # 8 X 7 Dimensional array with second lower diagonal
```

Author [\[deleted\]](#)Difficulty [Easy](#)

Max Score 20

Submitted By [114892](#)

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Task

Your task is to print an array of size **N**×**M** with its main diagonal elements as **1**'s and **0**'s everywhere else.

Note

In order to get alignment correct, please insert the line `numpy.set_printoptions(legacy='1.13')` below the numpy import.

Input Format

A single line containing the space separated values of **N** and **M**.

N denotes the rows.

M denotes the columns.

Output Format

Print the desired $N \times M$ array.

Sample Input

3 3

Sample Output

```
[[ 1.  0.  0.]
 [ 0.  1.  0.]
 [ 0.  0.  1.]]
```

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Language

Pypy 3



1
2
3
4
5
6
7
8
9
10
11

Line: 11 Col: 1



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